RS2 — The hypothesis that the singlet superfluid is the Higgs field, on the deflating 4+4 and 7+1 spacetimes

PROJECT spacetimes, Revision_Deflation. Supplement to both manuscripts of the revision

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0. Purpose and result

The author's task states the hypothesis, called **H2** in the main project, that the superfluid singlet state — the condensate of the fermion pair along the hidden axis $x0$ — *is the Higgs field*. The supplement S2 of the main project tests the hypothesis against seven conditions, C1 to C7, that a Higgs field must meet, on the two flat spacetimes. This supplement repeats the test on the two spacetimes of the revision, in which $x5$, $x6$, $x7$ deflate. It is written for a reader who has S2 at hand; the conditions are restated in Sec. 1.

Result. The deflation changes the standing of the hypothesis in three ways.

1. *What supported the hypothesis on the flat 7+1 spacetime is weakened.* There, the condensate breaks the rotation group $SO(4)$ of the four extra space axes to $SO(3)$, the pattern of the Higgs vacuum, and the $SO(4)$ is a symmetry of the spacetime. On the deflating 7+1 spacetime the rotations that mix $x0$ with a deflating coordinate are no longer symmetries of the spacetime. The internal symmetry survives, exactly, as a symmetry of the fields that do not depend on the extra coordinates; but it is now a *hidden* symmetry, as it was all along on 4+4 (Sec. 3).
2. *The condensate in the vacuum* (condition C2), open on flat spacetime because it needs a coupling α_{VS} of about 2 or more, is reached by the deflation at a definite time — but outside the range in which the model can be trusted (Sec. 4).
3. *A new and severe condition appears.* If the Higgs field is the pair condensate, the electroweak scale depends on the coupling α_{VS}, and α_{VS} grows with the deflation. The masses of the ordinary fermions would then change with time, and atomic clocks exclude this unless the present rate of deflation is below about 10^{-6} of the Hubble constant (Sec. 5). **The hypothesis H2 and a deflation that is of any consequence today exclude each other.**

0.1 How to reproduce

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
./code/deflation-rs/target/release/deflation symmetry    # Secs. 2 and 3
./code/deflation-rs/target/release/deflation cosmology   # Sec. 5 (check def-C6)
python3 code/scripts/make_figures.py isometries bounds
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py docs_src/RS2_higgs_hypothesis_deflating.src.md
bash code/scripts/build_docs.sh supplement/RS2_higgs_hypothesis_deflating.md
```

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1. The hypothesis and the conditions

In the Standard Model the Higgs field is a doublet of the weak-isospin group with a vacuum value $v_{\mathrm{EW}} = 246.22$ GeV, which breaks the electroweak symmetry, gives the W and Z bosons and the quarks and charged leptons their masses, and leaves one scalar particle of 125 GeV. Its four real components form a vector of a group $SO(4)$, which the vacuum value breaks to $SO(3)$. S2 derives from this the conditions that a condensate must satisfy in order to *be* the Higgs field:

	Condition
C1	it is a scalar under the Lorentz group of ordinary spacetime
C2	it is non-zero in empty space
C3	it is a doublet of the weak-isospin group with a neutral component
C4	it breaks $SO(4)$ to $SO(3)$
C5	its scale is 246 GeV, with a scalar particle of 125 GeV with the observed couplings
C6	its couplings give the quarks and leptons their masses
C7	no further light or charged particles accompany it that would have been seen

Two statements that must be kept apart.

- **H2** is the author's hypothesis: the *pair condensate* is the Higgs field. The electroweak scale is then the scale of the condensate.
- **HV** is a different assumption, used in one place of the supplement RS3: the vacuum value of the *Vorton-Scalar* sets the electroweak scale. S2 already notes that the Vorton-Scalar is a singlet of all non-Abelian symmetries and cannot be the Higgs field of the Standard Model. HV is not examined here.

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2. What the deflation leaves unchanged: conditions C1, C3 and C4 in the sector

The pair operators, their transformation properties and the internal symmetry are properties of the Dirac matrices of the orthonormal frame, which the deflation does not change, and of the equations of the fields that do not depend on the extra coordinates, which for the weighted fields are the flat equations (DMD, Sec. 3.3). The program computes the internal symmetry of this sector again, for both deflating spacetimes, with the routine of the main project (check `def-S3`; the 24 items are listed in Appendix B). Every item has the value that S2 and S1 found:

- the matrices that commute with the dynamics form the algebra $su(2)_A \oplus su(2)_B \oplus u(1)_F \oplus u(1)_E$, of dimension 8;
- the eight internal states of a fermion are the two doublets $(2,1) + (1,2)$;
- the four vector pair operators along the frame axes of $x0, x5, x6, x7$ form the $(2,2)$ with fermion number 2 — the representation of a Higgs field — and their one invariant quadratic form is the frame metric of the four axes;
- a condensate along one axis breaks $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$ to $SO(3)$.

Hence, *within the sector*:

C1 holds as on flat spacetime: the pair along the hidden axis is invariant under the rotations of $x1, x2, x3$, which remain symmetries of the spacetime, and under the boosts of ordinary spacetime as they act on the fields of the sector. (The boosts are no longer symmetries of the deflating spacetime as a whole, DMD, Part 5; in the sector this shows only in the dependence of the couplings on the time.)

C3 and C4 hold in the form in which S2 found them for the zero-mode sector of 4+4: the pattern is right, under the internal symmetry of the sector.

3. What the deflation changes: the symmetry is no longer a symmetry of the spacetime

3.1 The count

On the flat 7+1 spacetime the six generators of $SO(4)$ are the rotations of the six pairs of axes taken from $x0, x5, x6, x7$: symmetries of the spacetime. This is what S2 counts in favour of the hypothesis for 7+1 — a symmetry of the extra dimensions is the kind of symmetry from which a gauge interaction can arise, as in the theories of Kaluza and Klein. The program tests each of the 28 rotations and boosts of each flat spacetime on the deflating spacetime, for three choices of the rates (checks `44d-S1`, `71d-S1`, `44d-S2`, `71d-S2`):

spacetime	rotations and boosts	isometries, rates (1, 2, 3)	rates (2, 2, 5)	rates (1, 1, 1)	internal generators	of these isometries: (1, 2, 3)	(2, 2, 5)	(1, 1, 1)
7+1	28	6	7	9	6	0	1	3
4+4	28	6	7	9	6	0	1	3

(The rates in the heading are the values of $\mathsf{g55}, \mathsf{g66}, \mathsf{g77}$ used for three different, two equal and three equal rates. "Internal generators" are the six rotations of the pairs from $x0, x5, x6, x7$.) Of the six internal generators,

- **none** is a symmetry of the spacetime if the three rates differ;
- **one**, the rotation of the two coordinates with the same rate, if two rates are equal;
- **three**, the rotations of $x5, x6, x7$ among themselves, if the three rates are equal.

The three rotations that mix the hidden axis $x0$ with a deflating coordinate are never symmetries of the deflating spacetime: $x0$ keeps its size and the other three shrink.

3.2 The meaning for the 7+1 model

For equal rates the symmetry of the *spacetime* is $SO(3)$, the rotations of the three deflating coordinates — precisely the subgroup that a condensate along $x0$ would leave unbroken. The deflation thus breaks $SO(4)$ to $SO(3)$ *explicitly*, at the level of the geometry, in the same pattern in which the hypothesis needs the condensate to break it *spontaneously*. Three consequences follow.

1. The three broken generators, which under H2 would correspond to the massive weak bosons, are not symmetries of the spacetime at all. A gauge interaction for them cannot arise from the geometry of the deflating spacetime.
2. In the sector of fields that do not depend on the extra coordinates the full $SO(4)$ nevertheless acts as an exact symmetry (Sec. 2). It is a hidden symmetry: a symmetry of the equations of a sector that is not a symmetry of the spacetime. S2 found exactly this situation on the flat 4+4 spacetime and concluded that the hypothesis has no geometric basis there. **The deflation puts the 7+1 model in the position of the 4+4 model.**
3. The potential of the order parameter does not single out the hidden axis (RS1, Sec. 5.2): within the sector the direction of the condensate is as free as on flat spacetime. Outside the sector the axes differ — a variation of the order parameter along a deflating coordinate costs a gradient energy that grows as $1/b_t^2$ — so that the hidden symmetry is exact only for fields that are strictly independent of the extra coordinates.

3.3 The meaning for the 4+4 model

On the flat 4+4 spacetime the only non-Abelian internal symmetry that the geometry provides is $SO(3)_T$, the rotations of the three additional times, and under it the hidden-axis condensate is a

singlet (S2, Sec. 4.2). On the deflating 4+4 spacetime $SO(3)_T$ is a symmetry of the spacetime only if the three rates are equal, and is reduced to one rotation, or to none, otherwise. The conclusion of S2 for 4+4 is unchanged where the rates are equal and holds a fortiori where they are not: the hypothesis has no geometric basis in 4+4.

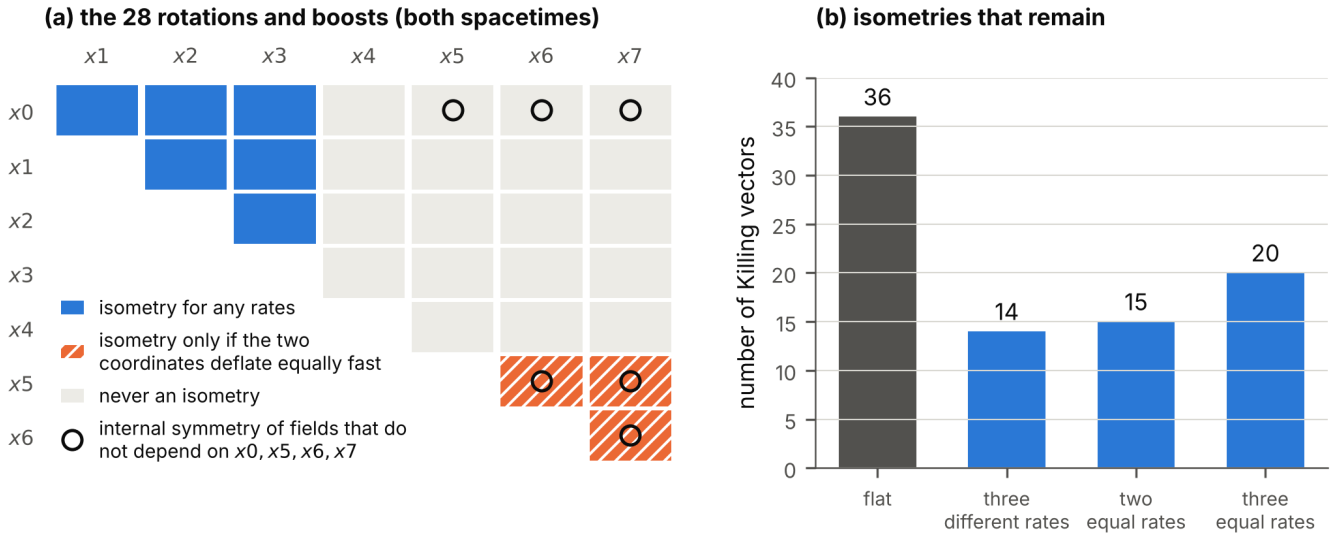


Figure 1: The 28 rotations and boosts on the deflating spacetimes, and the number of Killing vectors (the figure of DMD, Sec. 1.5). The circles mark the six internal generators: for three different rates none of them is a symmetry of the spacetime. Data: `data/isometries.csv`; program: `deflation symmetry`.

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4. Condition C2: a condensate in empty space

S2 leaves C2 open: by the ladder equation of the supplement S4 a pair condensate forms in the vacuum only if α_{VS} exceeds a critical value α_c of about 2 or more, depending on the cutoff. On a deflating spacetime α_{VS} grows as e^{τ_d} and reaches any such value at a definite time. The supplement RS4, Sec. 5, shows what else has happened by then: for α_{VS} of order one the one-boson exchange is no longer a controlled approximation, and the field at which the one-loop potential of the Vorton-Scalar falls below its vacuum value has moved to within a factor of two of the vacuum value. **C2 can be reached by waiting, but not within the range of validity of the model.** The status of C2 remains “open”.

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5. A new condition: the electroweak scale must not drift

5.1 The argument

Suppose H2 holds. The electroweak scale v_{EW} is then the scale of the pair condensate in the vacuum. That scale is determined by the parameters of the model: the masses m and m_s, the cutoff, and the coupling α_{VS}. On the deflating spacetime the masses are constant and $\alpha_{VS} \propto e^{\tau_d}$ (RS1, Sec. 1.3). Therefore

$$v_{\mathrm{EW}} \propto e^{n_{\mathrm{EW}}\tau_d}, \qquad n_{\mathrm{EW}} = \frac{d\ln v_{\mathrm{EW}}}{d\ln \alpha_{VS}}. \tag{5.1}$$

The exponent cannot vanish: below the critical coupling there is no condensate, so the scale depends on the coupling, and most strongly just above the critical coupling.

What follows for ordinary matter. Under H2 the masses of the electron and of the quarks are proportional to v_{EW} and change at the rate n_{EW} gsum. The mass of the proton is due mostly to the strong interaction and does not follow. The ratio m_p/m_e of the two masses therefore changes at the rate n_{EW} gsum. Comparisons of atomic clocks find this ratio constant to 8.0×10^{-17} per year at two sigma (Ref. [1]). Hence

$$\boxed{\text{gsum} < \frac{8.0 \times 10^{-17}\ \text{per year}}{n_{\mathrm{EW}}} = \frac{1.16 \times 10^{-6}\, H_0}{n_{\mathrm{EW}}}} \tag{5.2}$$

(check `def-C6`). The bound needs no assumption about gravity. For $n_{\mathrm{EW}} = 1$ it is 2787 times stronger than the bound that lunar laser ranging gives if gravity is eight-dimensional (RS3, Sec. 3).

5.2 The size of the exponent

For the model itself — massive fermions and a massive radial mode — the scale of the vacuum condensate has not been computed as a function of the coupling. It is known in closed form for the case on which S4 tests its ladder equation: massless fermions and massless bosons between an infrared scale and a cutoff. There the scale of the condensate obeys

$$\varrho_c \,\ln\frac{\text{cutoff}}{\text{scale}} = \arctan\frac{1}{2\varrho_c}, \qquad \varrho_c^2 = \frac{1}{4}\Big(\frac{\alpha_{VS}}{\alpha_c} - 1\Big),$$

(the relation of S4, Sec. 7.2, written with the ratio of the coupling to its critical value; ϱ_c is only an abbreviation for the square root on the right), from which

$$n_{\mathrm{EW}} = \frac{(1 + 4\varrho_c^2)\ \arctan(1/2\varrho_c)}{8\,\varrho_c^3} + \frac{1}{4\varrho_c^2}. \tag{5.3}$$

The program verifies Eq. (5.3) against the numerical derivative to 3×10^{-8} and tabulates it, together with the bound (5.2):

α_{VS}/α_c	ln(cutoff/scale)	n_{EW}	bound on gsum (per year)	bound in units of H_0
1.05	12.082	147	5.4×10^{-19}	7.9×10^{-9}
1.10	7.998	54.0	1.5×10^{-18}	2.2×10^{-8}
1.25	4.429	15.1	5.3×10^{-18}	7.7×10^{-8}
1.50	2.702	6.05	1.3×10^{-17}	1.9×10^{-7}
2.00	1.571	2.57	3.1×10^{-17}	4.5×10^{-7}
3.00	0.870	1.15	6.9×10^{-17}	1.0×10^{-6}
5.00	0.464	0.540	1.5×10^{-16}	2.2×10^{-6}
10.0	0.215	0.230	3.5×10^{-16}	5.0×10^{-6}

The exponent diverges at the critical coupling, equals $1 + \pi/2 = 2.57$ at twice the critical coupling, and exceeds 1 for $\alpha_{VS} < 3.27\,\alpha_c$. It falls below 1 only where the scale of the condensate approaches the cutoff, that is, where the four-dimensional description ends. *Within the range in which a condensate of the kind that H2 needs can be described at all, n_{EW} is of order one or larger.*

5.3 The conclusion

By Eq. (5.2), under H2 the deflation clock can have advanced by at most gsum $t_0 < 1.1 \times 10^{-6}/n_{\mathrm{EW}}$ during the age t_0 of the universe at a constant rate. Every effect of the deflation described in the supplements RS1, RS3 and RS4 is then unobservably small. Conversely, a deflation at any rate that those supplements find interesting excludes H2.

This conclusion does not depend on the unresolved conditions C2 and C5 to C7. It depends on two properties only: that the electroweak scale is set by the dynamics of the model, and that the coupling of the model follows the volume of the deflating coordinates. The first is the content of H2; the second is exact (RS1, Sec. 1).

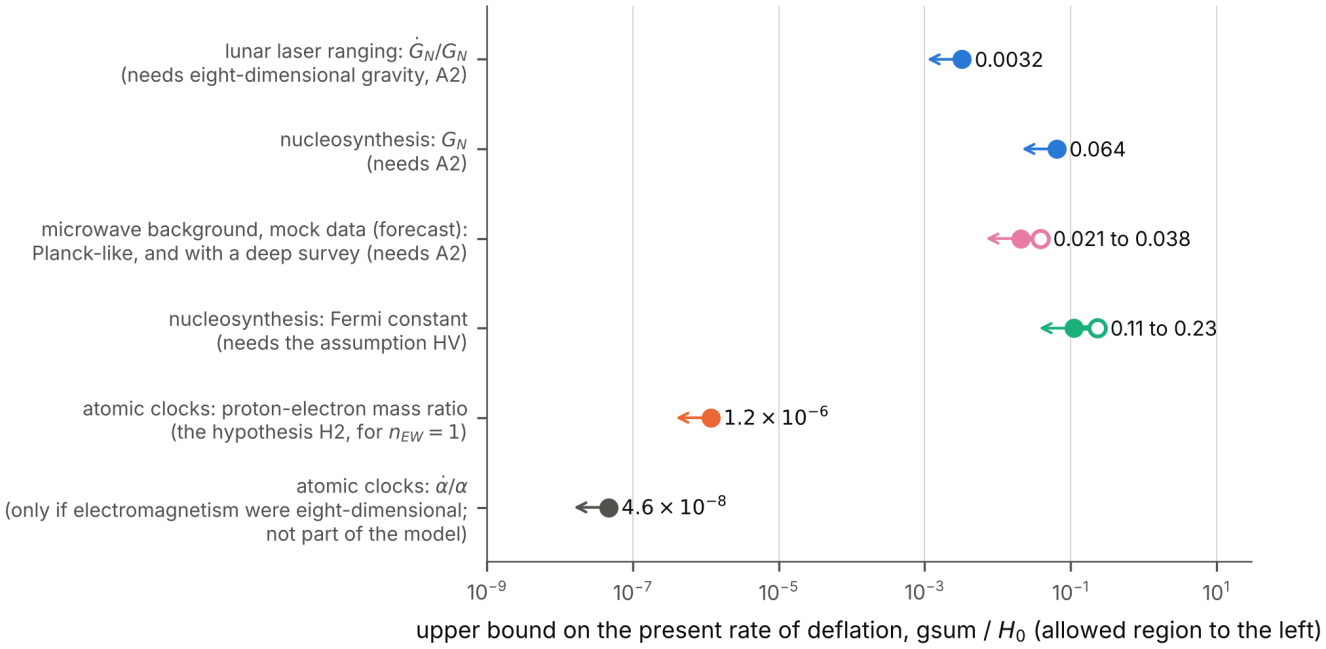


Figure 2: Upper bounds on the present rate of deflation (the figure of RS3, Sec. 3). The bound that follows from the hypothesis H2 and the atomic clocks, Eq. (5.2) with $n_{\mathrm{EW}} = 1$, is the fourth from the top. Data: `data/deflation_bounds.csv`; program: `deflation cosmology`.

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6. Verdict

Condition	deflating 7+1	deflating 4+4	Basis
C1 scalar in ordinary spacetime	met in the sector	met in the sector	Sec. 2; check `def-S3`
C2 condensate in the vacuum	open; reached in time only outside the range of validity	the same	Sec. 4; RS4, Sec. 5
C3 doublet with a neutral component	only under a hidden symmetry (three different rates), or with $SO(3)$ alone as a symmetry of the spacetime (equal rates)	only under a hidden symmetry; singlet under $SO(3)_T$	Sec. 3; checks `71d-S2`, `44d-S2`
C4 breaking $SO(4) \to SO(3)$	met in the sector; the spacetime itself has at most the $SO(3)$	met in the sector only	Secs. 2, 3
C5 scale and scalar mass	unfavourable (estimate of S2, unchanged)	—	S2, Sec. 5
C6 masses of ordinary fermions	not addressed	—	S2, Sec. 5
C7 no unobserved particles	additional particles predicted	—	S2, Sec. 5
new: the electroweak scale does not drift	requires gsum $< 1.16 \times 10^{-6} H_0/n_{\mathrm{EW}}$	the same	Sec. 5; check `def-C6`

What has been established.

1. The internal symmetry of the sector of fields that do not depend on the extra coordinates, the representation of the pair operators and the breaking pattern are unchanged by the deflation (Sec. 2).
2. Of the six generators of that symmetry, the deflating spacetime keeps none, one or three as symmetries of the spacetime; the three that mix the hidden axis with a deflating coordinate are never symmetries (Sec. 3). On the deflating 7+1 spacetime the structural point that S2 counted in favour of the hypothesis survives only for a hidden symmetry.
3. Under H2 the electroweak scale follows the coupling with the exponent n_{EW}, and atomic clocks bound the rate of deflation by Eq. (5.2). n_{EW} is given in closed form for the scale-invariant ladder equation (Sec. 5).

What has not been established.

1. Everything that S2 lists as not established: that a vacuum condensate forms, that rotations of extra dimensions are the electroweak gauge group, the scale and the mass of the scalar, the masses of ordinary fermions.
2. The exponent n_{EW} for the model itself. Eq. (5.3) holds for massless fermions and bosons; the statement that n_{EW} is of order one or larger rests on that case and on the general fact that the scale vanishes at the critical coupling.
3. Whether a mechanism exists that would make the electroweak scale insensitive to the coupling of the pair. None is known in the model.

The hypothesis “the singlet superfluid is the Higgs field” is therefore, on the deflating spacetimes, **unproven, with less in its favour than on the flat 7+1 spacetime, and incompatible with any deflation that is fast enough to matter today.**

Appendix A. Output of the programs

A.1 `deflation symmetry`

```
PROJECT spacetimes, Revision_Deflation: checks of the symmetries on the deflating spacetimes
 ↪  (isometries, internal symmetry of the fields that do not depend on the extra coordinates, zero
 ↪  modes on a vortex line, reduction of the couplings)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation symmetry
18 of 18 checks passed

[PASS] 44d-S1  (exact) of the 28 rotations and boosts of the flat 4+4 spacetime (one for each pair
 ↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
 ↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
 ↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
 ↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
 ↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
 ↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
 ↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 44d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
 ↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
 ↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
 ↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
 ↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
 ↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
 ↪  symmetries)
```

```
[PASS] 71d-S1  (exact) of the 28 rotations and boosts of the flat 7+1 spacetime (one for each pair
↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 71d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
↪  symmetries)
[PASS] def-S3  (numerical) the internal symmetry of the fields that do not depend on the extra
↪  coordinates, computed again for both spacetimes (24 items of data/reduced_symmetry.csv, all
↪  as expected): the commutant of the five matrices has dimension 8 and is u(2) + u(2) = su(2)_A
↪  + su(2)_B + u(1)_F + u(1)_E; the eight one-particle states are (2,1) + (1,2); the four vector
↪  pair operators are the (2,2) with fermion number 2, and their invariant quadratic form is the
↪  frame metric of the axes x0, x5, x6, x7. The deflation does not change this symmetry, because
↪  the weighted field of this sector obeys the flat equations (checks 44d-D.., 71d-D.. of the
↪  command matrices)
[PASS] 44d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
↪  one and anticommute with one another
[PASS] 44d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
↪  square one
[PASS] 44d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 44d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
↪  the generator P of U(1)_E
[PASS] 44d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 44d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] 71d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
↪  one and anticommute with one another
[PASS] 71d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
↪  square one
[PASS] 71d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 71d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
↪  the generator P of U(1)_E
[PASS] 71d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 71d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] def-S4  (exact) reduction to fields that do not depend on x5, x6, x7: sqrt|g| times the
↪  Yukawa term and the potential of the Vorton-Scalar, written with the weighted fields psihat =
↪  sqrt(Vd) psi and Phihat = sqrt(Vd) Phi_VS, are the same expressions with the couplings
↪  y_VS/sqrt(Vd), lambda_VS/Vd and the vacuum value v_VS sqrt(Vd); the products y_VS v_VS and
↪  lambda_VS v_VS^2, hence the fermion mass and the mass of the radial mode, do not change
↪  (rational sample values, exact). The exponents of data/coupling_exponents.csv follow, with Vd
↪  proportional to exp(-gsum x4)
```

```
[PASS] ALL  (exact) every check above passed
```

A.2 deflation cosmology

```
PROJECT spacetimes, Revision_Deflation: checks of the cosmological calculations on the deflating
 ↪  spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation cosmology
6 of 6 checks passed

[PASS] def-C1  (numerical) without deflation the background integrals of this module agree with
 ↪  the main project (its recombination calculation gives z_* = 1089.92): 100 theta_* = 1.03965
 ↪  here against 1.03965, sound horizon 144.449 Mpc against 144.449 Mpc; age 13.814 Gyr
[PASS] def-C2  (numerical) with deflation at a constant rate, gravity was weaker in the past, the
 ↪  early universe expanded more slowly at a given density, and the sound horizon is larger;
 ↪  holding 100 theta_*, omega_b and omega_c fixed, the inferred Hubble constant FALLS
 ↪  monotonically with the rate: d ln H_0/d(gsum/H_0) = -0.969 at small rates; H_0 = 66.70, 65.36,
 ↪  60.37 km/s/Mpc for gsum = 0.01, 0.03, 0.1 H_0 (67.36 without deflation). Deflation moves the
 ↪  Hubble constant away from the locally measured value. These numbers come from the full
 ↪  calculation of recombination on the deflating background (last scattering moves from z_* =
 ↪  1089.92 to 1088.04 at gsum = 0.1 H_0, and the helium fraction falls). If z_* and the helium
 ↪  fraction are held at their values without deflation, the slope is -0.911 and the values are
 ↪  66.74, 65.48, 60.77 km/s/Mpc: that approximation underestimates the fall by 5.8 to 6.0 per
 ↪  cent
[PASS] def-C3  (numerical) bounds on the present rate of deflation if gravity is eight-dimensional
 ↪  (G_N grows at the rate gsum): lunar laser ranging, (dG_N/dt)/G_N = (7.1 +- 7.6) 10^-14 per
 ↪  year, gives gsum < 2.23e-13 per year = 3.24e-3 H_0 at two sigma; nucleosynthesis,
 ↪  G_N(nucleosynthesis)/G_N(today) > 0.94 at two sigma, gives gsum < 4.43e-12 per year = 0.064
 ↪  H_0 for a rate that was constant since then. At these rates the inferred Hubble constant
 ↪  changes by -0.211 and -4.38 km/s/Mpc, in the direction away from the local measurement, 73.50
 ↪  +- 0.81 km/s/Mpc. Over the age t_0 of the model without deflation (H_0 t_0 = 0.9516) the
 ↪  deflation clock has advanced by gsum t_0 = 3.1e-3 at the first of these rates
[PASS] def-C5  (numerical) bounds that do not rest on gravity: (i) under the assumption HV (the
 ↪  vacuum value of the Vorton-Scalar sets the electroweak scale) the Fermi constant grows at the
 ↪  rate gsum while every mass stays constant; nucleosynthesis with fixed masses allows G_Fermi at
 ↪  t = 1 s to differ by 10 to 20 per cent from today's value (Scherrer and Spergel 1993), which
 ↪  gives gsum < 0.111 to 0.234 H_0, weaker than the bound from Newton's constant; (ii) in the
 ↪  model as it stands (without H2 and without HV) no mass of ordinary matter depends on the
 ↪  deflation, and the measured constancy of the ratio of the proton mass to the electron mass
 ↪  does not bound the rate; (iii) the fine-structure constant would bound it, gsum < 3.2e-18 per
 ↪  year = 4.6e-8 H_0, only if the electromagnetic field propagated along the deflating
 ↪  coordinates, which is not part of the model
[PASS] def-C6  (numerical) the author's hypothesis H2 (the superfluid state is the Higgs field) on
 ↪  a deflating spacetime: the electroweak scale is then the scale of the pair condensate, which
 ↪  depends on alpha_VS, and alpha_VS grows as exp(gsum x4). With n_EW = d ln v_EW/d ln alpha_VS
 ↪  the electron mass changes at the rate n_EW gsum and the proton mass does not; atomic clocks
 ↪  (ratio of the two masses constant to 8.0e-17 per year at two sigma) then give gsum < 8.0e-17
 ↪  per year / n_EW = 1.16e-6 H_0 / n_EW, a bound 2787 times stronger than that of lunar laser
 ↪  ranging for n_EW = 1, and one that needs no assumption about gravity; over the age of the
 ↪  universe the deflation clock can then have advanced by at most 1.1e-6/n_EW. In the
 ↪  scale-invariant ladder equation of the main project n_EW = (1 + 4 rho_c^2) atan(1/(2
 ↪  rho_c))/(8 rho_c^3) + 1/(4 rho_c^2) with rho_c^2 = (alpha_VS/alpha_c - 1)/4 (agreement with
 ↪  the numerical derivative: 2.8e-8): n_EW = 15.1, 2.57, 1.15, 0.54 for alpha_VS/alpha_c = 1.25,
 ↪  2, 3, 5; it diverges at the critical coupling and exceeds 1 for alpha_VS < 3.27 alpha_c. For
 ↪  the model itself (massive fermions, massive radial mode) n_EW has not been computed
```

```
[PASS] def-C4  (numerical) dark matter of bound pairs loses rest mass while the coupling grows,
 ↪  because the binding energy increases (four examples with alpha_VS = 0.4 to 1 today; binding
 ↪  energy at most 8.2 per cent of the rest mass). If gravity feels the deflation, the inferred
 ↪  Hubble constant changes by between -4.11 and -0.19 km/s/Mpc at the rates that lunar laser
 ↪  ranging and nucleosynthesis allow: it falls, the weaker gravity of the past outweighing the
 ↪  loss of mass. (The runs with a varying rest mass hold the redshift of last scattering and the
 ↪  helium fraction at their values without deflation, which by def-C2 underestimates a fall by
 ↪  about six per cent of itself.) If Newton's constant is held fixed and the deflation acts on
 ↪  the dark sector alone, rates between 0.03 and 1 H_0 change the inferred Hubble constant by at
 ↪  most +0.48 km/s/Mpc, for a loss of up to 7.7 per cent of the rest mass
 ↪  (data/hubble_mass_loss.csv). The loss of rest mass is the only effect of the deflation that
 ↪  raises the inferred Hubble constant; in these examples it supplies less than a tenth of the
 ↪  6.1 km/s/Mpc that separate 67.36 from the local value 73.50
[PASS] ALL  (numerical) every check above passed
```

Appendix B. The internal symmetry of the sector, item by item

The 24 items of `data/reduced_symmetry.csv`, as the program writes them (check `def-S3`). The symbols in this table are those of the main registry: P is the generator of $U(1)_E$, T_i are the generators of the two ideals $su(2)_A$ and $su(2)_B$, and $C^{-}\gamma^{a}$ are the matrices of the four vector pair operators.

item	7+1	4+4	expected
dimension of the commutant, momenta along x1 x2 x3	8	8	8
dimension of the commutant, momenta along x0 x1 x2 x3	4	4	4
dimension of the commutant, momenta along every space coordinate of the model	1	4	1 (7+1); 4 (4+4)
dimension of the centre of the commutant	2	2	2
|P^2 - 1| for the traceless central generator P	0.00e0	0.00e0	0
dimensions of the two ideals	[3, 3]	[3, 3]	[3, 3]
structure constants |f_123| of the ideals (generators with Tr T_i T_j = 2 delta_ij)	1.000000000000 1.000000000000	1.000000000000 1.000000000000	1 1 (su(2))
|sum_i T_i^2 - (3/4) projector| of the ideals	0.00e0 0.00e0	2.22e-16 2.22e-16	0 0 (doublets)
positive-energy states with P = +1 and P = -1 at a test momentum	4.000000 4.000000	4.000000 4.000000	4 4: (2,1) + (1,2)
closure of the four vector pair operators C^-gamma^a, a in {0,5,6,7}	0.00e0	3.14e-16	0
|Casimir - 3/4| of the two ideals on the four pair operators	0.00e0 0.00e0	5.55e-17 5.55e-17	0 0: the (2,2)
fermion number of the pair operators	2.000000	2.000000	2
charge of the pair operators under the central generator P	0.00e0	0.00e0	0
number of invariant quadratic forms on the four pair operators	1	1	1
the invariant form (diagonal; frame axes 0 5 6 7; first entry 1)	+1.000000 +1.000000 +1.000000 +1.000000	+1.000000 -1.000000 -1.000000 -1.000000	+1 +1 +1 +1 (7+1); +1 -1 -1 -1 (4+4): the frame metric of the extra axes up to sign
largest off-diagonal entry of the invariant form	0.00e0	2.22e-16	0
complex dimension of the particle-antiparticle mixing generators, no chemical potential	2	2	2 (symmetry O(4) x O(4), dimension 12)
the same with a chemical potential	0	0	0 (symmetry U(2) x U(2), dimension 8)

item	7+1	4+4	expected
gamma^0 gamma^5	antihermitian; |[.,dynamics]| = 0.00e0	Hermitian; |[.,dynamics]| = 0.00e0	antihermitian (7+1: rotation); Hermitian (4+4: not a boost); commutes
gamma^0 gamma^6	antihermitian; |[.,dynamics]| = 0.00e0	Hermitian; |[.,dynamics]| = 0.00e0	antihermitian (7+1: rotation); Hermitian (4+4: not a boost); commutes
gamma^0 gamma^7	antihermitian; |[.,dynamics]| = 0.00e0	Hermitian; |[.,dynamics]| = 0.00e0	antihermitian (7+1: rotation); Hermitian (4+4: not a boost); commutes
gamma^5 gamma^6	antihermitian; |[.,dynamics]| = 0.00e0	antihermitian; |[.,dynamics]| = 0.00e0	antihermitian (rotation) in both; commutes
gamma^5 gamma^7	antihermitian; |[.,dynamics]| = 0.00e0	antihermitian; |[.,dynamics]| = 0.00e0	antihermitian (rotation) in both; commutes
gamma^6 gamma^7	antihermitian; |[.,dynamics]| = 0.00e0	antihermitian; |[.,dynamics]| = 0.00e0	antihermitian (rotation) in both; commutes

Appendix C. Files

file	content	written by
`code/deflation-rs/src/symmetry_defl.rs`, `cmd_symmetry.rs`	test of the 28 rotations and boosts; internal symmetry of the sector	—
`data/isometries.csv`	Sec. 3	`deflation symmetry`
`data/reduced_symmetry.csv`	Sec. 2 and Appendix B	`deflation symmetry`
`data/higgs_scale_exponent.csv`, `data/deflation_bounds.csv`	Sec. 5	`deflation cosmology`
`data/symmetry_checks.txt`, `data/cosmology_checks.txt` (and `.csv`)	Appendix A	the two commands
`figures/fig_isometries`, `figures/fig_bounds` (`.pdf`, `.png`)	the two figures	`make_figures.py`

References

1. R. Lange, N. Huntemann, J. M. Rahm, C. Sanner, H. Shao, B. Lipphardt, Chr. Tamm, S. Weyers and E. Peik, *Improved limits for violations of local position invariance from atomic clock comparisons*, Phys. Rev. Lett. **126**, 011102 (2021).
2. The references of the supplement S2 of the main project.